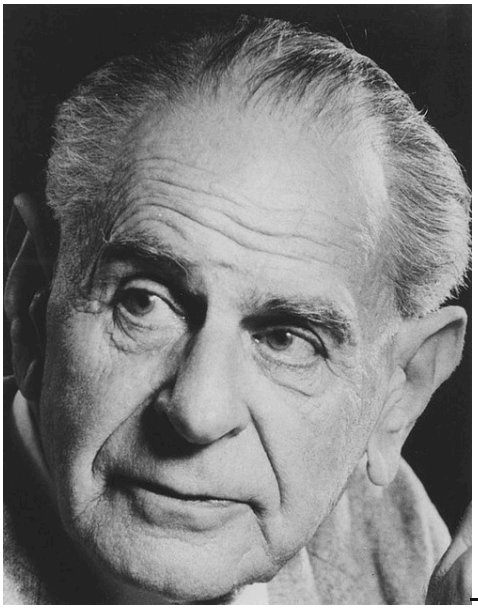
Karl Popper: The Line Between Science and Pseudoscience

It's not immediately clear to the layman what the essential difference is between science and something masquerading as science: pseudoscience (https://fs.blog/cargo-cult-science/). The distinction between the two is at the core of what comprises human knowledge: How do we actually know something to be true?

Sir **Karl Popper** (1902-1994), the scientific philosopher (https://en.wikipedia.org/wiki/Karl_Popper), was interested in the same problem. How do we actually define the scientific process? How do we know which theories can be said to be truly explanatory?


(https://www.farnamstreetblog.com/wp-content/uploads/2016/01/3833724834_397c34132c_z.jpg)

He began addressing it in a lecture, which is printed in the book *Conjectures and Refutations: The Growth of Scientific Knowledge (https://amzn.to/41BCv4l)*:

> *When I received the list of participants in this course and realized that I had been asked to speak to philosophical colleagues I thought, after some hesitation and consultation, that you would probably prefer me to speak about those problems which interest me most, and about those developments with which I am most intimately acquainted. I therefore decided to do what I have never done before: to give you a report on my own work in the philosophy of science, since the autumn of 1919 when I first began to grapple with the problem,* ***'When should a theory be ranked as scientific?' or 'Is there a criterion for the scientific character or status of a theory?'***

Privacy - Terms

Popper saw a problem with the number of theories he considered *non-scientific*. Theories that, on their surface, seemed to have a lot in common with good, hard, rigorous science. But the question of how we decide which theories are compatible with the scientific method, and those which are not, was harder than it seemed.

What's Scientific?

It is most common to say that science is done by collecting observations and grinding out theories from them. Charles Darwin said (https://fs.blog/charles-darwin-thinker/), after working long and hard on the problem (see the work of thinking (https://fs.blog/how-to-think/)):

> *My mind seems to have become a kind of machine for grinding general laws out of large collections of facts.*

This is a popularly accepted notion. We observe, observe, and observe, and we look for theories to best *explain* the mass of facts. (Although even this is not really true: Popper points out that we must start with some *a priori* knowledge to be able to generate new knowledge. Observation is always done with some hypotheses in mind — we can't understand the world from a totally blank slate.)

The problem, as Popper saw it, is that some bodies of knowledge, more properly named pseudoscience, would be considered scientific if the "Observe & Deduce" operating definition were left alone. For example, a believing astrologist can ably provide you with "evidence" that their theories are sound. The biographical information of a great many people can be explained this way, they'd say.

The astrologist would tell you, for example, about how "Leos" seek to be the centre of attention; ambitious, strong, seeking the limelight. As proof, they might follow up with a host of real-life Leos: World-leaders, celebrities, politicians, and so on. In some sense, the theory would hold up. The observations could be explained by the theory, which is how science works, right?

Popper lived during a time when psychoanalytic theories were all the rage at just the same time Einstein was laying out a new foundation for the physical sciences with the concept of relativity. What made Popper uncomfortable were *comparisons between the two*. Why did he feel so uneasy putting Marxist (https://fs.blog/the-effect-of-scale-on-values/) theories and Freudian psychology (https://amzn.to/3ILKmnv) in the same

category of knowledge as *Einstein's Relativity (https://amzn.to/3Y8kFTG)*? Did all three not have vast explanatory power in the world? Each theory's proponents certainly believed so, but Popper was not satisfied.

> *It was during the summer of 1919 that I began to feel more and more dissatisfied with these three theories–the Marxist theory of history, psychoanalysis, and individual psychology; and* ***I began to feel dubious about their claims to scientific status****. My problem perhaps first took the simple form, 'What is wrong with Marxism, psycho-analysis, and individual psychology? Why are they so different from physical theories, from Newton's theory, and especially from the theory of relativity?'*
>
> *I found that those of my friends who were admirers of Marx, Freud, and Adler, were impressed by a number of points common to these theories, and especially by their apparent explanatory power.* ***These theories appeared to be able to explain practically everything that happened within the fields to which they referred.*** *The study of any of them seemed to have the effect of an intellectual conversion or revelation, opening your eyes to a new truth hidden from those not yet initiated. Once your eyes were thus opened you saw confirming instances everywhere: the world was full of verifications of the theory.*
>
> ***Whatever happened always confirmed it.*** *Thus its truth appeared manifest; and unbelievers were clearly people who did not want to see the manifest truth; who refused to see it, either because it was against their class interest, or because of their repressions which were still 'un-analysed' and crying aloud for treatment.*

The Importance of Falsifiability

Here was the salient problem: The proponents of these new sciences saw validations and verifications of their theories everywhere. If you were having trouble as an adult, it could *always* be explained by something your mother or father had done to you when you were young, some repressed something-or-other that hadn't been analyzed and solved. They were confirmation bias (https://fs.blog/confirmation-bias/) machines.

What was the missing element? Popper had figured it out before long: **The non-scientific theories could not be *falsified*. They were not *testable* in a legitimate way. There was no possible objection that could be raised which would show the theory to be wrong.**

In a true science, the following statement can be easily made: "If *x* happens, it would show demonstrably that theory *y* is not true." We can then design an experiment, a physical one or sometimes a simple thought experiment, to figure out if *x* actually does

happen. It's the opposite of looking for verification; you must try to show the theory is *incorrect*, and if you fail to do so, thereby *strengthen* it.

Pseudosciences cannot and do not do this—they are not strong enough to hold up. As an example, Popper discussed Freud's theories of the mind (http://www.amazon.com/gp/product/067960166X/ref=as_li_qf_sp_asin_il_tl?ie=UTF8&camp=1789&creative=9325&creativeASIN=067960166X&linkCode=as2&tag=farnamstreet-20&linkId=CMT6RJQDMVH22UFU) in relation to Alfred Adler's so-called "individual psychology (http://www.amazon.com/gp/product/1578989841/ref=as_li_qf_sp_asin_il_tl?ie=UTF8&camp=1789&creative=9325&creativeASIN=1578989841&linkCode=as2&tag=farnamstreet-20&linkId=ENE46IVNFYR3HMHC)," which was popular at the time:

> *I may illustrate this by two very different examples of human behaviour: that of a man who pushes a child into the water with the intention of drowning it; and that of a man who sacrifices his life in an attempt to save the child.* ***Each of these two cases can be explained with equal ease in Freudian and in Adlerian terms.*** *According to Freud the first man suffered from repression (say, of some component of his Oedipus complex), while the second man had achieved sublimation. According to Adler the first man suffered from feelings of inferiority (producing perhaps the need to prove to himself that he dared to commit some crime), and so did the second man (whose need was to prove to himself that he dared to rescue the child).* ***I could not think of any human behaviour which could not be interpreted in terms of either theory****. It was precisely this fact–that they always fitted, that they were always confirmed–which in the eyes of their admirers constituted the strongest argument in favour of these theories.* ***It began to dawn on me that this apparent strength was in fact their weakness.***

Popper contrasted these theories against Relativity, which made specific, verifiable predictions, giving the conditions under which the predictions *could be shown false*. It turned out that Einstein's predictions came to be **true** when tested, thus verifying the theory through attempts to falsify it. But the essential nature of the theory gave grounds under which it could have been wrong. To this day, physicists seek to figure out where Relativity *breaks down* in order to come to a more fundamental understanding of physical reality. And while the theory may eventually be proven incomplete or a special case of a more general phenomenon, it has still made accurate, testable predictions that have led to practical breakthroughs.

Thus, in Popper's words, **science requires testability**: *"If observation shows that the predicted effect is definitely absent, then the theory is simply refuted."* This means a good theory must have an element of **risk** to it. It must be able to be proven wrong under stated conditions.

Popper's Essential Conclusions

From there, Popper laid out his essential conclusions, which are useful to any thinker trying to figure out if a theory they hold dear is something that can be put in the scientific realm:

1. It is easy to obtain confirmations, or verifications, for nearly every theory–if we look for confirmations.

2. Confirmations should count only if they are the result of risky predictions; that is to say, if, unenlightened by the theory in question, we should have expected an event which was incompatible with the theory–an event which would have refuted the theory.

3. Every 'good' scientific theory is a prohibition: it forbids certain things to happen. The more a theory forbids, the better it is.

4. A theory which is not refutable by any conceivable event is nonscientific. Irrefutability is not a virtue of a theory (as people often think) but a vice.

5. Every genuine test of a theory is an attempt to falsify it, or to refute it. Testability is falsifiability; but there are degrees of testability: some theories are more testable, more exposed to refutation, than others; they take, as it were, greater risks.

6. Confirming evidence should not count except when it is the result of a genuine test of the theory; and this means that it can be presented as a serious but unsuccessful attempt to falsify the theory. (I now speak in such cases of 'corroborating evidence'.)

7. Some genuinely testable theories, when found to be false, are still upheld by their admirers–for example by introducing ad hoc some auxiliary assumption, or by re-interpreting the theory ad hoc in such a way that it escapes refutation. Such a procedure is always possible, but it rescues the theory from refutation only at the price of destroying, or at least lowering, its scientific status. (I later described such a rescuing operation as a 'conventionalist twist' or a 'conventionalist stratagem'.)

One can sum up all this by saying that the criterion of the scientific status of a theory is its falsifiability, or refutability, or testability.

Finally, Popper was careful to say that it is impossible to prove that Freudianism was *not true*, at least in part. But we *can* say that we simply don't *know* whether it's true because it does not make **specific testable predictions**. It may have many kernels of truth in it, but we can't tell. This is the essential **"line of demarcation**," as Popper called it, between science and pseudoscience.

READ NEXT

Montaigne's Rule for Reading: The Promiscuous Pursuit of Pleasure

Michel de Montaigne (1533-1592) might have been the original essayist. George Orwell before George Orwell. Montaigne was well-read, smart, …

→